

Assignment_05CT

September 10, 2026

1 Assignment -05

2 Kushagra Singh 2401PH07

3 Q1

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[111]: import matplotlib.pyplot as plt
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[108]: import math
import matplotlib.pyplot as plt

X = [0.1, 0.2, 0.3, 0.4]
Y = [9.96664442, 4.93315488, 3.23272814, 2.36522242]

n = len(X)
DT = [[0 for j in range(n)] for i in range(n)]

for i in range(n):
    DT[i][0] = Y[i]

for j in range(1, n):
    for i in range(n-j):
        DT[i][j] = (X[i+j] - X[j-1]) / (DT[i+1][j-1] - DT[0][j-1])

print("Inverse Difference Table")

for i in range(n):
    for j in range(n-i):
        print(f"{DT[i][j]:12.6f}", end=" ")
    print()

x_values = []

x = 0.01
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while x <= 0.4:
    x_values.append(x)
    x = x + 0.001

y = []
e_rational = []

for x in x_values:

    p = DT[0][n-1]

    for i in range(n-2, -1, -1):
        p = DT[0][i] + (x-X[i])/p

    actual = math.cos(x)/math.sin(x)

    error = actual-p

    y.append(p)
    e_rational.append(error)


A = []
e_lagrange = []

for x in x_values:

    p = 0

    for i in range(4):

        l = 1

        for j in range(4):

            if i != j:
                l = l*(x-X[j])/(X[i]-X[j])

        p = p + l*Y[i]

    actual = math.cos(x)/math.sin(x)

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error = actual-p

A.append(p)
e_lagrange.append(error)

print("X Values:", x_values)
print("Rational Interpolated Values:", y)
print("Lagrange Interpolated Values:", A)
plt.plot(x_values, y, label="Rational Interpolation")
plt.plot(x_values, A, label="Lagrange Interpolation")
plt.plot(X, Y, 'o', label="Data Points")

plt.grid()
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Rational vs Lagrange Interpolation")
plt.legend()
plt.show()

```

Inverse Difference Table

9.966644	-0.019867	-10.169354	-2.852511
4.933155	-0.029700	-10.204411	
3.232728	-0.039466		
2.365222			

X Values: [0.01, 0.011, 0.012, 0.013000000000000001, 0.014000000000000002, 0.015000000000000003, 0.016000000000000004, 0.017000000000000005, 0.018000000000000006, 0.019000000000000006, 0.020000000000000007, 0.021000000000000008, 0.022000000000000001, 0.023000000000000001, 0.024000000000000001, 0.025000000000000012, 0.026000000000000013, 0.027000000000000014, 0.028000000000000014, 0.029000000000000015, 0.030000000000000016, 0.031000000000000017, 0.032000000000000015, 0.033000000000000015, 0.034000000000000016, 0.035000000000000002, 0.036000000000000002, 0.037000000000000002, 0.038000000000000002, 0.039000000000000002, 0.040000000000000002, 0.041000000000000002, 0.042000000000000002, 0.043000000000000024, 0.044000000000000025, 0.045000000000000026, 0.046000000000000003, 0.047000000000000003, 0.048000000000000003, 0.049000000000000003, 0.050000000000000003, 0.051000000000000003, 0.052000000000000003, 0.053000000000000003, 0.054000000000000034, 0.055000000000000035, 0.056000000000000036, 0.057000000000000004, 0.058000000000000004, 0.059000000000000004, 0.060000000000000004, 0.061000000000000004, 0.062000000000000004, 0.063000000000000004, 0.064000000000000004, 0.065000000000000004, 0.066000000000000004, 0.067000000000000005, 0.068000000000000005, 0.069000000000000005, 0.070000000000000005, 0.071000000000000005, 0.072000000000000005, 0.073000000000000005, 0.074000000000000005, 0.075000000000000005, 0.076000000000000005, 0.077000000000000005, 0.078000000000000006, 0.079000000000000006, 0.080000000000000006,

[illegible]

[illegible]

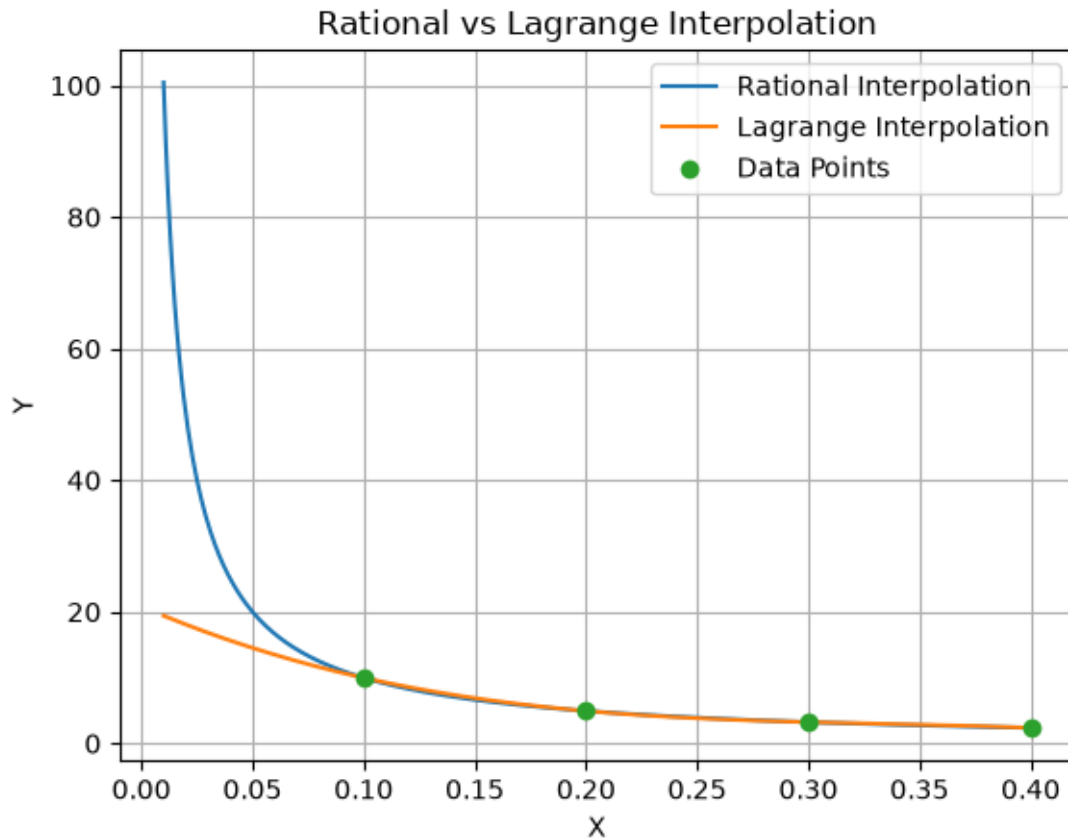
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2.4152923437043157, 2.403016195396006, 2.3905808254370364, 2.377983733685626]



```
[109]: plt.figure()
plt.plot(x_values, y)
plt.plot(X, Y, 'o')
plt.grid()
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Rational Interpolation")
plt.show()
```

```
plt.figure()
plt.plot(x_values, A)
```

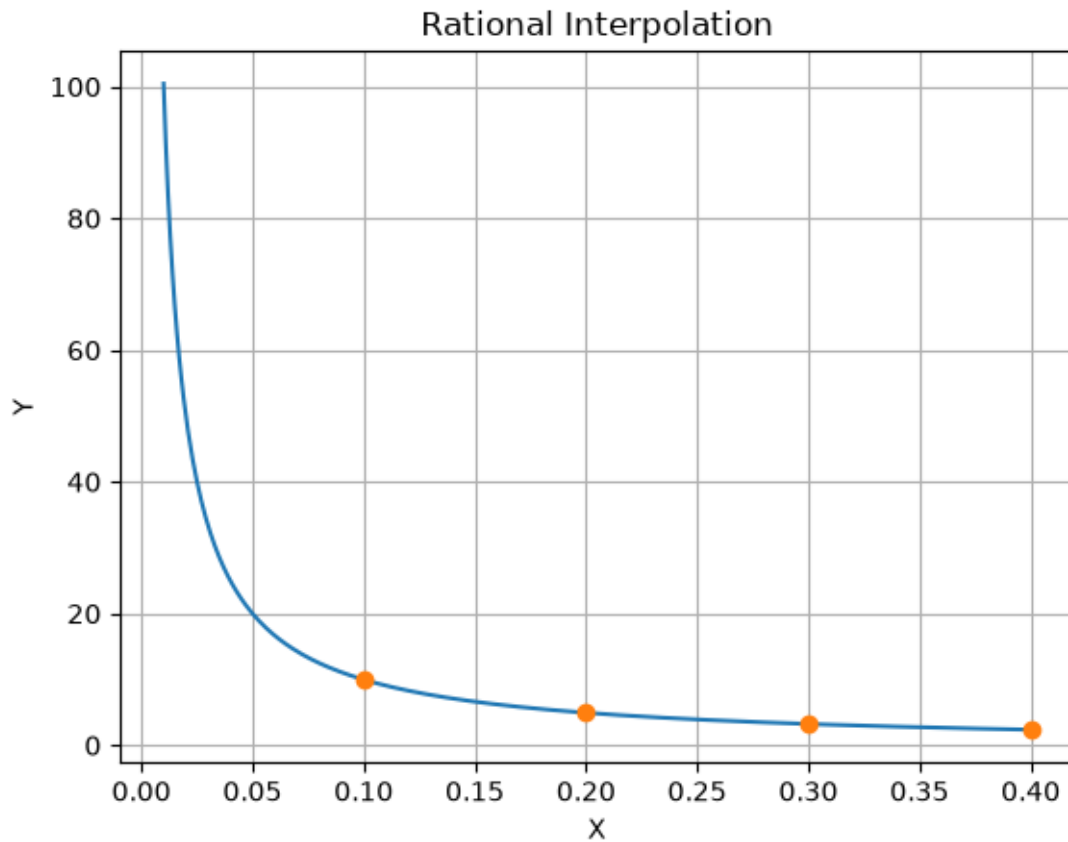
```

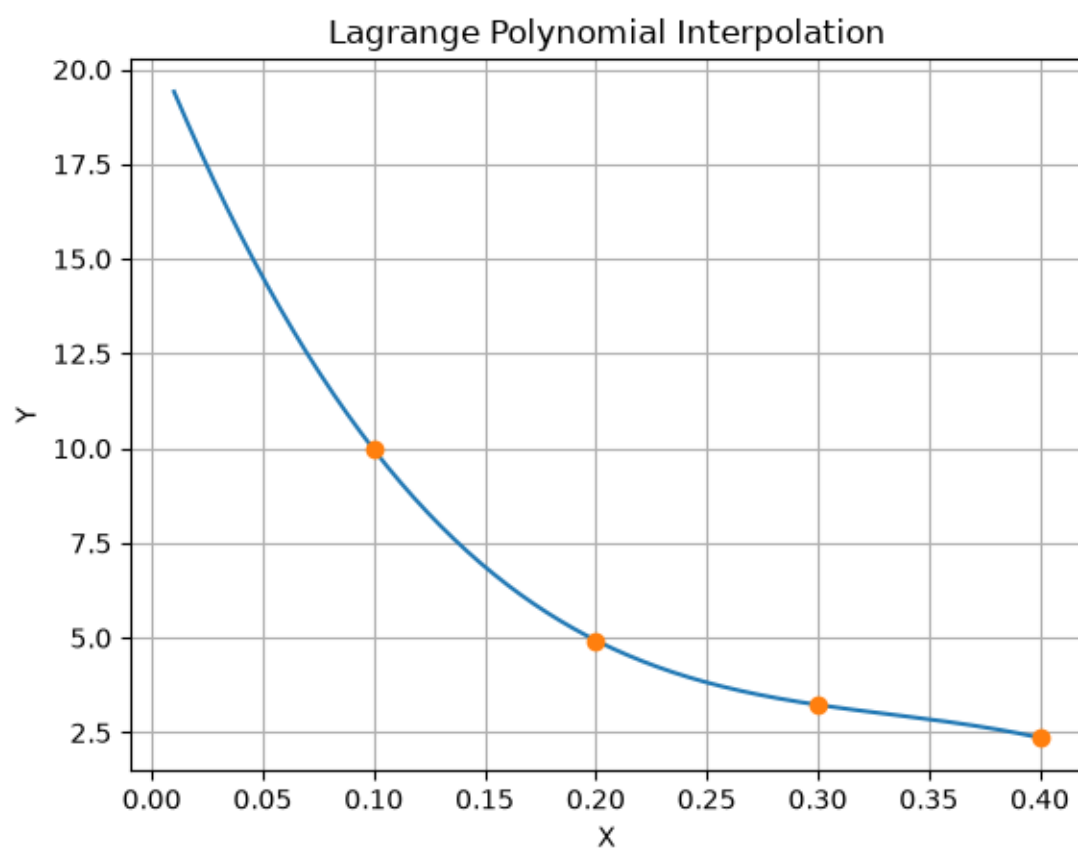
plt.plot(X, Y, 'o')
plt.grid()
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Lagrange Polynomial Interpolation")
plt.show()

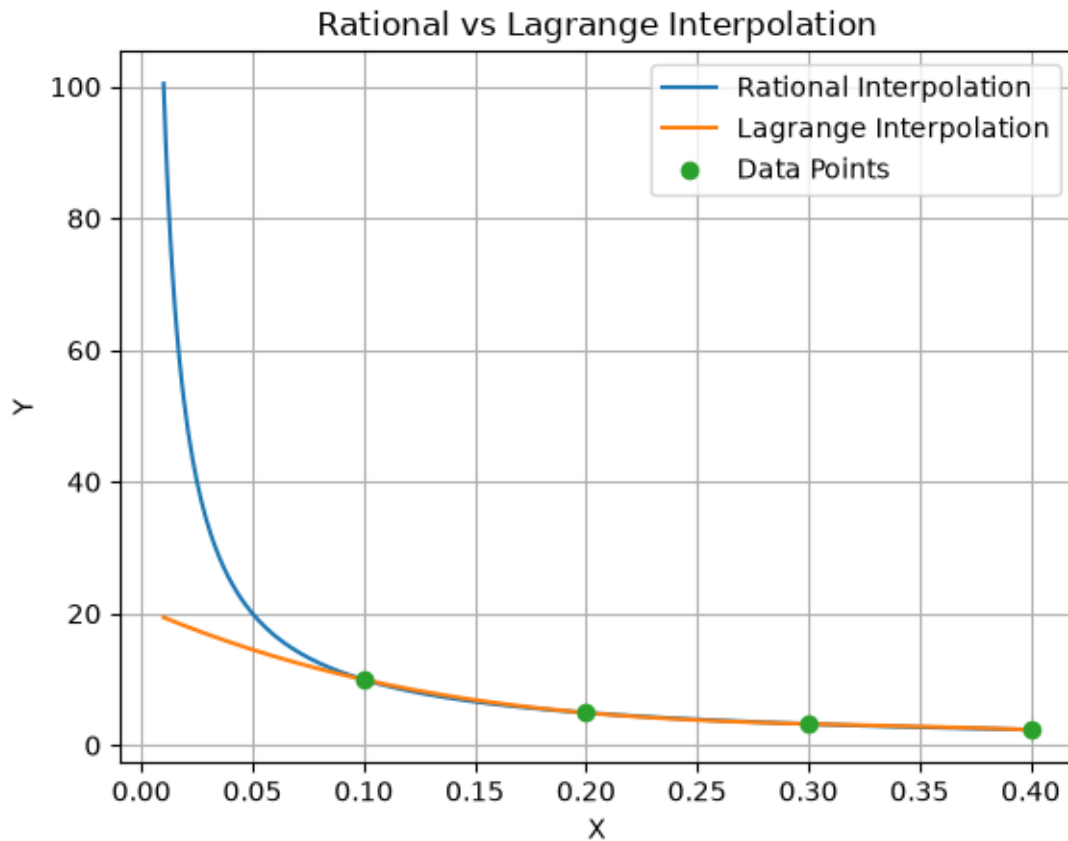
plt.figure()
plt.plot(x_values, y, label="Rational Interpolation")
plt.plot(x_values, A, label="Lagrange Interpolation")
plt.plot(X, Y, 'o', label="Data Points")

plt.grid()
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Rational vs Lagrange Interpolation")
plt.legend()
plt.show()

```







```
[84]: actual_values = []

for x in x_values:
    actual = math.cos(x) / math.sin(x)
    actual_values.append(actual)

plt.figure(figsize=(10, 6))

plt.plot(x_values, actual_values,
         label="Actual cot(x)",
         linestyle="--")

plt.plot(x_values, y,
         label="Rational Interpolation",

         markersize=3,
         linewidth=1)
```

```

plt.plot(x_values, A,
        label="Lagrange Interpolation",

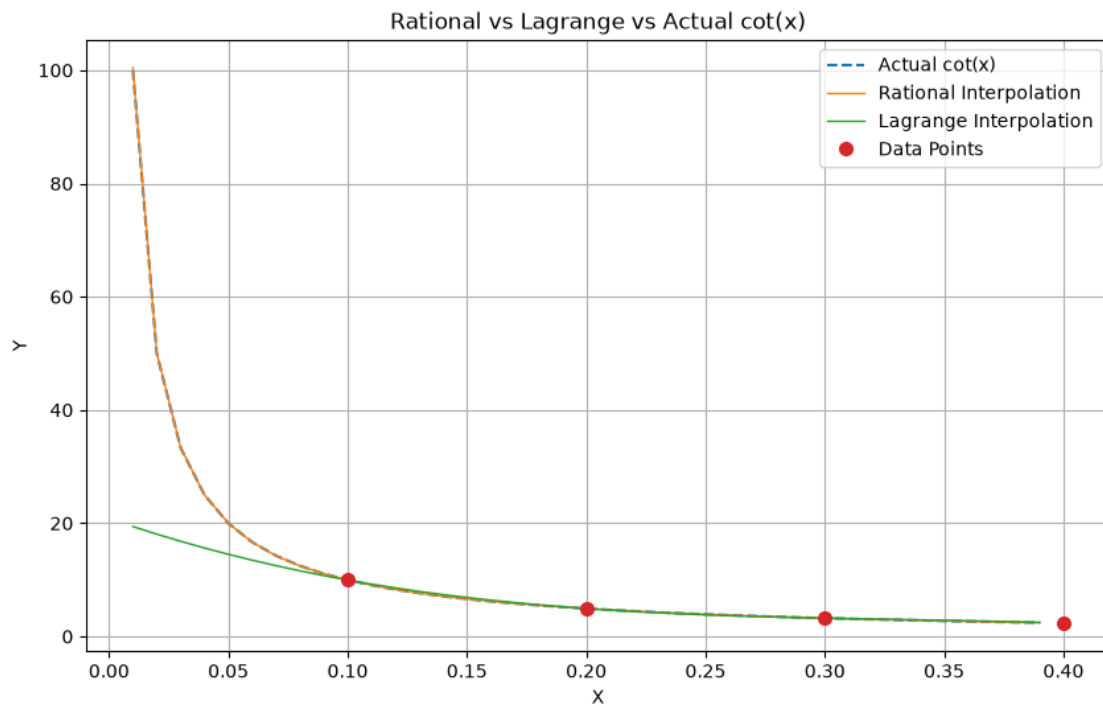
        markersize=3,
        linewidth=1)

plt.plot(X, Y, 'o',
        markersize=7,
        label="Data Points")

plt.grid()
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Rational vs Lagrange vs Actual cot(x)")
plt.legend()

plt.show()

```



4 Q2

```
[110]: import math
import matplotlib.pyplot as plt

X = [0, math.pi/2, math.pi]
Y = [0, 1, 0]

X_values = []
actual_error = []
error_bound = []

x = 0

while x <= math.pi:

    # Lagrange interpolation
    P = 0

    for i in range(3):
        L = 1

        for j in range(3):
            if i != j:
                L = L * (x - X[j]) / (X[i] - X[j])

        P = P + Y[i] * L

    E = abs(math.sin(x) - P)

    # Theoretical error bound

    bound = abs(x * (x - math.pi/2) * (x - math.pi)) / math.factorial(3)

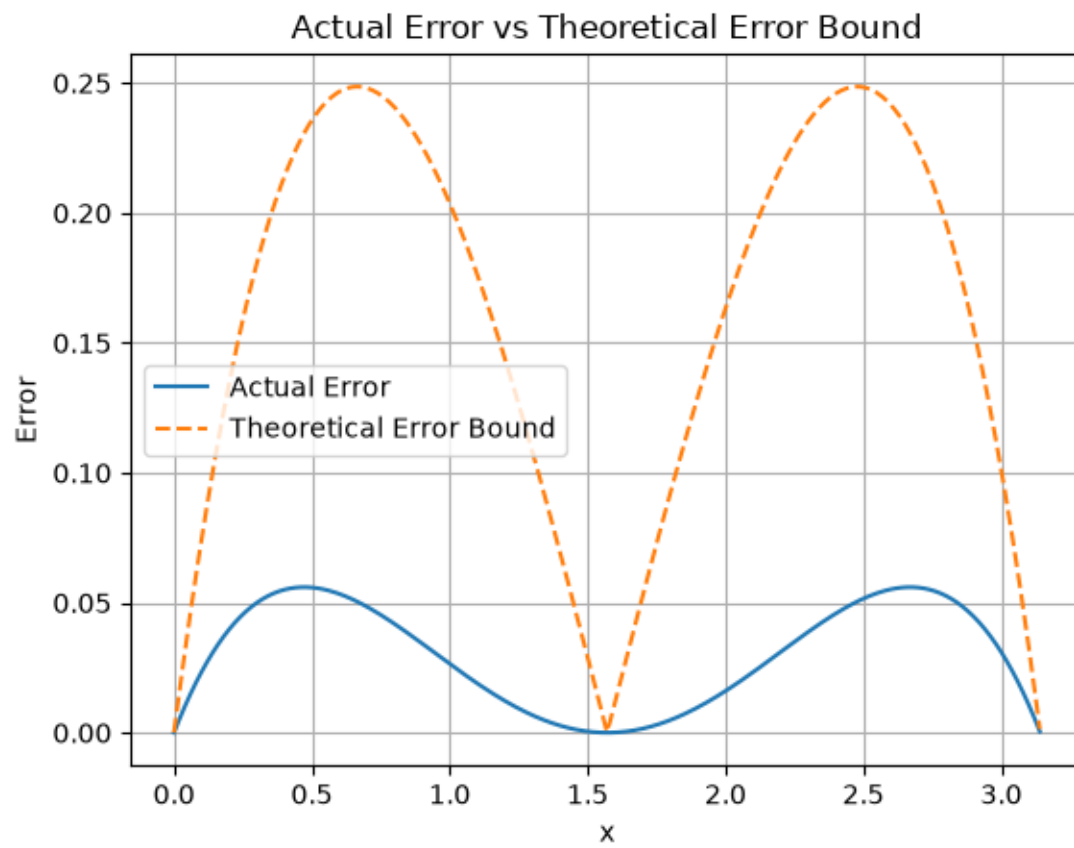
    X_values.append(x)
    actual_error.append(E)
    error_bound.append(bound)

    x = x + 0.01

plt.plot(X_values, actual_error, label="Actual Error")
plt.plot(X_values, error_bound, '--', label="Theoretical Error Bound")

plt.xlabel("x")
```

```
plt.ylabel("Error")
plt.title("Actual Error vs Theoretical Error Bound")
plt.legend()
plt.grid()
plt.show()
```



[]: